and developmental pattern. The function of the wings of a bat, the wings of a bird and wings of an insect are similar, but their basic structures are different.

iii. **Vestigial organs:** The **degenerated** and **non-functional organs** of animals are called vestigial organs. The same organs are found to be well-developed and functional, in some of the related forms. Some of the vestigial organs in man are vermiform appendix, nictitating membrane, caudal vertebra, coccyx etc.

iv. **Atavism:** The **reappearance of ancestral characters** in some individuals is called atavism. e.g. Presence of rudimentary tail in new born babies, presence of thick hair on the human body.

19.2.2 Evidences from Embryology

The study of comparative embryology of different animals, supports the concept of evolution. The embryos from fish to mammals are similar in their early stages of development. The differentiation of their special characters appear in the later stages of development.

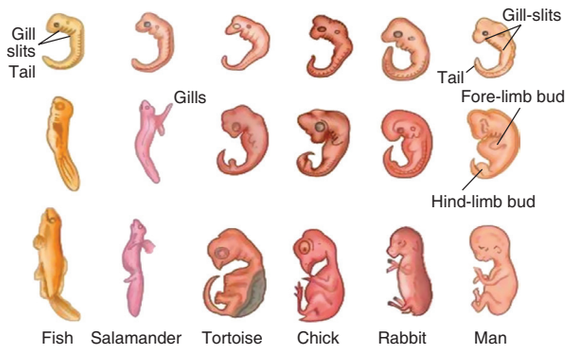


Figure 19.3 Embryonic development of vertebrates

Biogenetic law or **Recapitulation theory** was given by **Ernst Haeckel**. According to this theory, **Ontogeny recapitulates Phylogeny**. The stages of development of the individual animal repeats the evolutionary history of the entire race of the animal.

19.2.2 Evidences from Palaeontology

Palaeontology deals with the **study of fossils**. **Leonardo da Vinci** is called the **Father of Palaeontology**. The study of fossils helps us to understand the line of evolution of many invertebrates and vertebrates. Fossil records show that the evolution has taken a gradual process from simple to complex organisms. The origin of modern birds is supported by the evidences from palaeontology.

Archaeopteryx: Archaeopteryx is the oldest known **fossil bird**. It was an early bird-like form found in the Jurassic period.It is considered to be a **connecting link** between reptiles and birds. It had wings with feathers, like a bird. It had long tail, clawed digits and conical teeth, like a reptile.



Figure 19.4 Archaeopteryx - Fossil bird

19.3 Theories of Evolution

Life had evolved along with evolution of earth towards the end of 18th century. **Evolution** is the **gradual change** occurring in living organisms over a period of time. Formation of new species due to changes in specific characters over several generations as response to natural selection, is called evolution. The natural changes occuring is explained through the theories of evolution as proposed by Lamarck and Darwin.

10th_Science Unit-19.indd 276 12/3/2022 3:09:16 PM